MIL-A-27391B (USAF)
1 AUGUST 1979
SUPERSEDING
MIL-A-27391A (USAF)
18 April 1963

MILITARY SPECIFICATION

ALTIMETER, PRESSURE AAU-2/A

1. SCOPE

1.1 This specification covers the requirements for an integrally lighted pressure altimeter, designated AAU-2/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited)
PPP-B-636	- Box, Shipping, Fiberboard
PPP-B-601	- Boxes, Wood, Cleated Plywood

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

MIL-L-25467 - Lighting, Integral, Red, Aircraft Instrument, General Specification for
DOD-D-1000 - Drawings, Engineering and Associated List

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-143 - Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831 - Test Reports, Preparation of
MIL-STD-859 - Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-889 - Dissimilar Metals
MS28105 - Cover Glass, Aircraft Instrument Dial Glass
MS33558 - Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS27572 - Cases, Instrument, Flange-Mounted, Aircraft
MS33649 - Bosses, Fluid Connection - Internal Straight Thread

(Copies of specifications standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.5).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Fungus proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the altimeter except where magnetic materials are essential.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Corrosive fumes. Materials used in the interior construction of the altimeter shall not produce corrosive nor deleterious fumes under the test conditions specified herein.

3.3.6 Protective treatment. When materials are used that are subjected to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

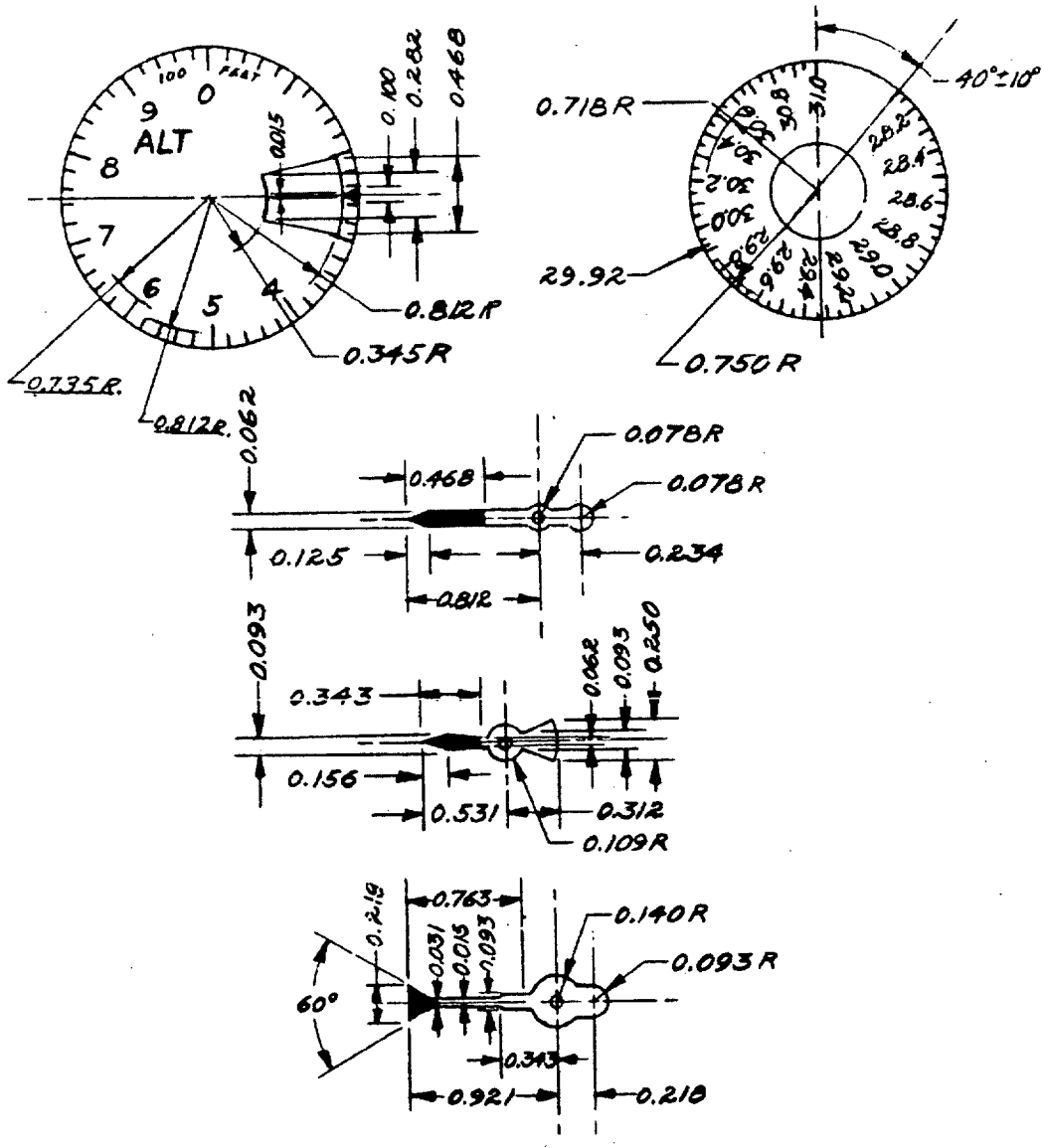
3.4 Design and construction.

3.4.1 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part number.

3.4.2 Altitude. The altitude display shall be in accordance with figure 1 and shall provide a visual display from 0 to 100,000 feet altitude. The altimeter shall be designed to withstand pressures equivalent to 80,000 feet altitude with no damage to the mechanism.

3.4.3 Transportability. The indicator shall be constructed to withstand failure to the normal strains, jars, vibrations and such other conditions as are incident to shipping, storage, installation, and service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ±0.010

FIGURE 1. Dials and pointers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

3.4.4 Barometric setting system. A barometric setting system, consisting of a knob and sump in the indicator, located in the lower left hand corner, shall be provided that will permit the altimeter to be set to indicate zero altitude at any existing ground level pressure throughout a range of 31 to 28.1 (inclusive) inches Hg. Suitable stops shall be provided in the barometric setting mechanism so that the scale cannot be read erroneously when the zero setting mechanism is moved beyond the range of 31 to 28.1 inches Hg. Rotation of the setting knob to either extreme shall not remove the numerals at each end of the scale completely from view.

3.4.4.1 Adjustment of setting. The zero setting system shall be provided with a suitable means of adjusting or correcting the relative setting of the altitude pointers and the zero setting system indications, should this become necessary. This adjustment shall be accessible from the face of the altimeter when installed on the instrument board and provided with a locking device in order to preclude the pilot's inadvertent adjustment of this setting instead of the zero altitude setting described in 3.4.4. Actuation of the locking device shall be accomplished without the necessity of either removing the altimeter from the instrument panel or removing the bezel from the altimeter.

3.4.4.1.1 Rotation of zero set. Clockwise rotation of the zero set knob shall cause increasing readings of altitude as indicated by the pointers as well as increasing numbers as indicated by the baro set subdial.

3.5 Crash provision. The altimeter shall be designed to facilitate reading of the instrument at the time of a crash provided there is no significant increase in item complexity or that reliability is not degraded (see 6.4).

3.6 Performance and environmental. The altimeter shall be capable of satisfactory operation when subjected to the following conditions:

a. Examination of product: The altimeter shall be inspected for conformance to specification requirements (see 4.6.1).

b. Altitude: Pressures ranging from 31 to .8099 inches Hg.

c. Scale errors: Scale errors shall not exceed the specified amount (see 4.6.2, 4.6.3 and 4.6.4).

d. Hysteresis: Hysteresis errors shall not exceed the specified amount (see 4.6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

e. Position: Position errors shall not exceed an indication of 20 feet (see 4.6.6).

f. Friction: Friction errors shall not exceed the specified amount (see 4.6.7).

g. Case leakage: Case leakage shall not exceed an indication of 100 feet (see 4.6.8).

h. Magnetic effect: Compass deflection shall exceed 1° in a magnetic field intensity of 0.17 to 0.19 oersted (see 4.6.9).

i. Temperature: Temperatures ranging from -62° to $+65^{o}$C (see 4.6.10, 4.6.11 and 4.6.12).

j. Humidity: Relative humidity up to 100 percent (see 4.6.13).

k. Salt spray: Exposure to simulated salt sea atmosphere (see 4.6.14).

l. Vibration:

(1) Vibration error shall not exceed a pointer variation of 20 feet when subjected to an amplitude of 0.003 to 0.005 inch at frequencies from 500 to 3000 cpm (see 4.6.15).

(2) The altimeter shall not fail when vibrated with a circular motion of 0.018 to 0.020 inch diameter at frequencies from 5 to 50 cps for a 3-hour period (see 4.6.16).

m. Fungus: Fungus growth as encountered in tropical climates (see 4.6.17).

n. Rain: Equivalent to four inches per hour (see 4.6.18).

o. Sand and dust: Exposure to sand and dust as encountered in desert areas (see 4.6.19).

p. Acceleration: Acceleration forces of 20g (see 4.6.20).

q. Endurance, stability and shelf life: When subjected to 2000 applications of 50,000 feet of altitude, 3 cycles of thermal shock, and 30 days of storage (see 4.6.21).

3.7 Case. The altimeter case shall conform to MS27572 for 2-inch dial except as shown on figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

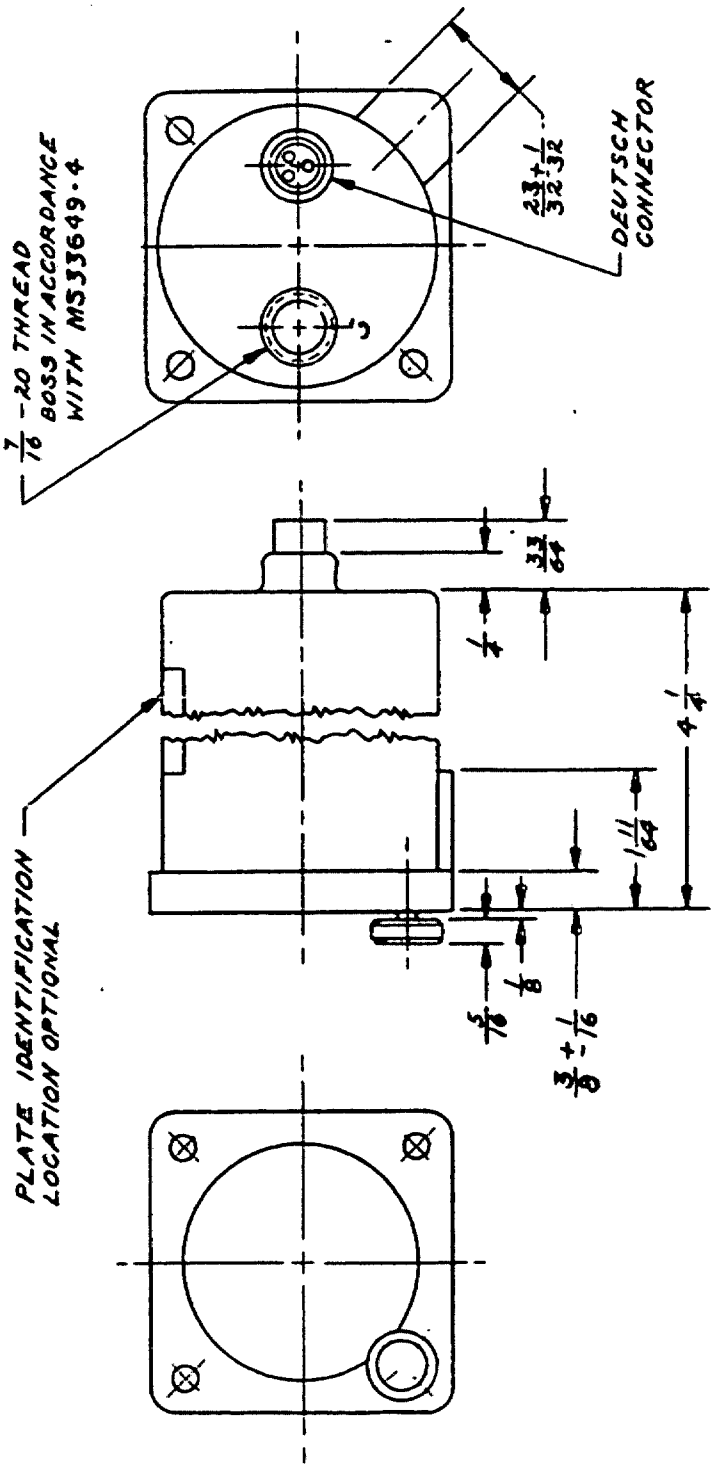


FIGURE 2. Case.

CASE IN ACCORDANCE WITH MS27572 EXCEPT AS SHOWN

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ± 1/64

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

3.7.1 Cover glass. The cover glass shall be in accordance with MS28105-7 and shall be properly sealed. It shall be possible to replace the cover glass by removing the bezel. The bezel shall be held in place by means of screws properly secured by lockwashers or similar devices of approved design. If the cover glass is used as part of the lighting system, a glass may be used with characteristics deviating from the glass specified herein to the extent required for the lighting function as long as it does not interfere with the readability of the instrument.

3.7.2 Dial to cover glass distance. The distance from dial to cover glass shall not be greater than 1/4 inch.

3.7.3 Shipping plug. Static pressure connection shall be provided with a vented plug. The plug shall incorporate a vent consisting of a hole 1/16 ±1/64 inch in diameter.

3.7.4 Case marking. The case, adjacent to the pressure connection, shall be marked as follows:

"S" - static connection

3.8 Integral lighting. Integral lighting shall be provided in accordance with MIL-L-25467. Lighting acceptability shall be based on a visual comparison with a prime standard. The methods of lighting shall be approved by the procuring activity prior to fabrication. The lighting shall operate on 5V dc.

3.8.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas, which cover the entire face of the indicator, shall be measured for brightness level in foot-lamberts. The prime standard instrument and its recorded overall brightness measurements shall be approved by the procuring activity. The standard shall be checked periodically, at least every 100 hours of operation, to determine any change in brightness. Variations that exceed ±20 percent in two or more areas shall require return of the standard for correction.

3.9 Dials.

3.9.1 Numerals and lettering. The dials shall conform to figure 1. The style and proportions of the numerals and lettering shall conform to MS33558. Numerals shall distinctly indicate the graduation to which they apply.

3.9.2 Markings. The markings listed below shall be finished in dull white, color No. 37875, of FED-STD-595.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

Main Dial	Height or Length Inch ±0.010	Width of Line or Graduation Inch ±0.010
Dial numerals 0, 1, 4, 5, 6, 7, 8, 9	0.125	0.020
Minor graduation	0.125	0.015
Major graduation	0.187	0.031
Subdial pointer	See figure 1	
Lettering "ALT"	0.140	0.020
Notation, 100 feet	0.078	0.010
Subdial		
Numerals	0.100	0.015
Major graduations	0.125	0.015
Minor graduations	0.093	0.010
29.92 mark	0.060	0.015

All markings not otherwise specified shall be finished in durable dull black, color No. 37038, of FED-STD-595.

3.9.3 Visibility of dial markings. All graduations, numerals, pointers, and other markings on the dial shall be readable from any point within the frustrum of a cone, the side of which makes an angle of not less than 20° with the perpendicular to the dial and the small diameter of which is the aperture of the dial.

3.10 Pointers. The pointers shall be proportioned in accordance with figure 1.

3.11 Electrical connections. Electrical connections shall conform to figure 3. The electrical connector on the case shall be a Deutsch Type DM5600-3P, or equal, located as shown on figure 2.

3.12 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight or loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free from backlash.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

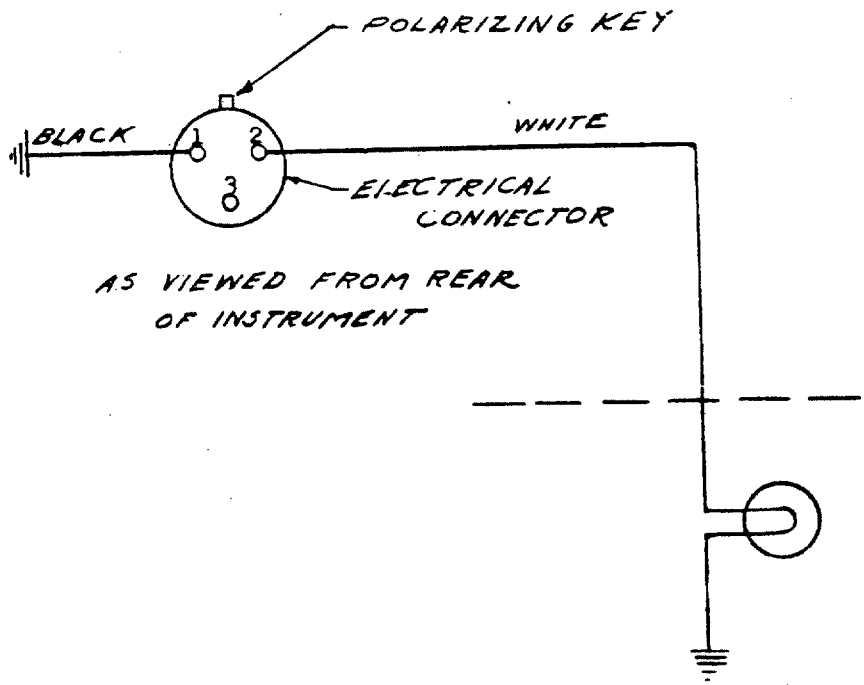


FIGURE 3. Wiring diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

3.14 Dimensions. Dimensions and tolerances that are not specified shall be as close as possible, consistent with the best shop practices. Where dimensions and tolerances may affect interchangeability, operation, or performance of the altimeter, they shall be held or limited accordingly.

3.15 Weight. The weight of the indicator completely assembled without connecting fittings shall not exceed 1 pound.

3.16 Finishes and protective coatings.

3.16.1 Anodic coatings. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that dials, small holes, inserts and parts in a hermetically sealed container need not be anodized.

3.16.2 Chemical coatings. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment shall be used in lieu of anodizing. Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing. When abrasion resistance is a factor, chemical films shall not be used in lieu of anodizing.

3.16.3 Tin plate. When the part is plated with tin over a copper flash, it need not be anodized or chemically coated.

3.16.4 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.17 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.18 Workmanship. The altimeter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, and freedom of parts from burrs and sharp edges. After final assembly, the altimeter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection and testing of the altimeter shall be classified as follows:

a. First article inspection

b. Quality conformance inspections

4.3 Inspection conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it shall be understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 300 and 3000 cpm. The altimeter shall be subjected to vibration in such manner that a point on the instrument case will describe, in a plane inclined 45° to the horizontal plane, a circle no less than 0.003 nor more than 0.005 inch diameter or a circle no less than 0.018 nor more than 0.020 inch diameter, as specified in the tests.

4.3.3 Tapping. The altimeter shall be tested in a normal operating position and shall be vibrated or tapped lightly before a test reading is taken (see 6.3.1).

4.3.4 Standard pressures. Standard pressures used in calibrating the altimeter shall be as specified in MIL-STD-859.

4.3.5 Attitude. Unless otherwise specified, tests shall be made with the indicator in a normal operating position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

4.3.6 External connections. When external connections are required to provide a simulated installation, they shall consist of standard AN or MS connections and at least 10 feet of 1/4 inch copper tubing that is coiled and arranged in such manner that it will provide drainage of condensed vapors. The connections shall be attached to the static pressure port of the indicator.

4.4 First article inspection. First article inspection shall consist of all tests described under 4.6, Methods of inspection.

4.4.1 Test sample. The test sample shall consist of three altimeters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 First article test report and test sample. When first article tests are conducted at a location other than the laboratory of the activity, the following shall be furnished to that activity:

a. Test report: Three copies of a test report in accordance with MIL-STD-831.

b. Test sample: The sample that was tested.

4.5 Quality conformance inspections.

a. Individual inspections

b. Sampling plan and inspections

4.5.1 Individual inspections. Each altimeter shall be subjected to the Individual tests of MIL-L-25467 and to the following inspections as described under 4.6, Methods of inspection:

a. Examination of product

b. Zero setting scale

c. Scale error at room temperature

d. Scale error at -35°C

e. Hysteresis

f. Position error

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

g. Friction

h. Case leakage

4.5.2 Sampling plans and inspections.

4.5.2.1 Sampling plan A. One altimeter, selected at random from each 100 or less produced on the contract or order, shall be subjected to the Individual inspections above, Sampling plan A tests of MIL-L-25467, and to the following inspections as described under 4.6, Methods of inspection:

a. Magnetic effect

b. High temperature

c. Vibration error

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first 15 items of the contract or order and shall be subjected to Sampling plan A above, Sampling plan B tests of MIL-L-25467 and the following tests as described in 4.6, Methods of inspections.

a. Low temperature exposure

b. High temperature exposure

c. Humidity

d. Salt spray

e. Vibration failure

f. Fungus

g. Rain

h. Sand and dust

i. Acceleration

j. Endurance, stability, and shelf life

4.5.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and corrected. The contractor shall explain to the Government representative the cause

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.4 Individual tests continued. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the entire items on hand or items procured later shall not be made until it is determined that all items meet all of the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Methods of inspection.

4.6.1 Examination of product. The altimeter shall be inspected to determine conformance to the requirements of this specification with respect to materials, workmanship, and marking.

4.6.2 Zero setting scale. The altimeter shall be subjected to atmospheric pressure. The altimeter shall be tapped or vibrated lightly and the pointer indication noted. After the series of readings have been recorded, the difference beteween the readings at the setting of 29.92 and every other setting shall be found. These differences shall be recorded and compared with the values in the "Correct Differences" column specified in table I. The recorded values shall not differ from the correct differences by more than 25 feet at any setting.

TABLE I. Zero setting mechanism

Setting Pressure Scale	Correct Difference
28.1	1,727
28.5	1,340
29.0	863
29.5	392
29.92	0
30.5	531
30.90	893
30.99	974

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

4.6.3 Scale error at room temperature. The zero setting system shall be set so that the pressure scale will read 29.92. While the altimeter is subjected to atmospheric pressure, the reading on the pointers shall be taken and the error of this indication determined by comparison with the barometer. Without changing the zero setting, the altimeter shall be subjected successively to the pressure specified in table II. The reduction in pressure, as indicated by a standard mercurial barometer, during the interval between test points, shall be made at a rate corresponding to an increase in altitude of approximately 3000 fpm. Before a test reading is taken, the altimeter shall remain at the pressure corresponding to each test point for no less than 1 nor more than 5 minutes. The scale errors shall not exceed the tolerances specified in table II. With no vibration applied to the altimeter, the movement of the pointer shall be observed for any irregular motion. From 0 to 25,000 feet, the pointer shall not stop nor jump more than 120 feet. From 25,000 to 40,000 feet, the pointer shall not stop nor jump more than 450 feet. From 40,000 feet to maximum range, the pointer shall not stop nor jump more than 850 feet. Wait one to five minutes, then return pointers to their original reading and correct for any change in atmospheric pressure, within the tolerance given in table III as the third test point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

TABLE II. Scale errors.

Standard Altitudes	Equivalent Pressure mm Hg	Room Temperature Variation From Standard Altitude	-35°C Variation From Room Temp Reading	-54° Variation From Room Temp Reading
0	760.00	±50	±50	±75
500	746.37	50		
1,000	732.93	70		
2,000	706.65	80		
3,000	681.14	90		
4,000	656.38	100		
6,000	609.05	110	100	150
8,000	564.51	120		
10,000	522.65	130		
12,000	483.34	150	175	270
14,000	446.46	175		
16,000	411.90	190		
18,000	379.53	210	275	420
20,000	349.25	225		
22,000	320.96	275		
25,000	282.03	300	375	560
30,000	225.69	325		
35,000	178.83	425	525	790
40,000	140.66	500		
45,000	110.62	625		
50,000	86.99	750	700	1500
55,000	68.40	875		
60,000	53.79	1000	1000	2250

TABLE III. Hysteresis scale error.

Test Points	Altitude	Equivalent Atmospheric Pressure	Permissible Variation
First point	25,000	281.9	150
Second point	20,000	349.1	150
Third point	----	Ground Level	60

4.6.4 Scale error at -35°C. The altimeter shall be maintained at -35° ±2°C for a 3-hour period and shall be tested while at this temperature. The test points and tolerances specified in table II shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

4.6.5 Hysteresis. Not more than 15 minutes after the altimeter has been subjected to the pressure corresponding to the upper limit of the scale in the test specified in 4.6.3, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3000 fpm until the pressure corresponding to the first test point given in table III is reached. The altimeter shall be remain at this pressure for at least 5 but not more than 15 minutes before the test reading is taken. After the reading has been taken, the pressure shall be increased at the specified rate until the pressure corresponding to the second test point given in table III is reached. The altimeter shall remain at this pressure for at least 1 but not more than 10 minutes before the test reading is taken. After the reading has been taken, the pressure shall be increased at the specified rate until atmospheric pressure is reached. The reading of the altimeter at either of the two test points shall not differ from the reading of the altimeter for the corresponding altitude in the test specified in 4.6.3 by more than the tolerance given in table III.

4.6.6 Position error. The reading shall be taken while the altimeter is in a vertical position (dial up) and after it has been vibrated. The altimeter shall then be rotated 90° to its normal operating position. Readings shall be taken after vibration in the 12, 3, 6, and 9 o'clock positions. None of the readings shall differ from the 12 o'clock reading by more than 20 feet. Sufficient vibration shall be applied to the unit to remove all of the friction from the instrument.

4.6.7 Friction. The altimeter shall be tested for friction at each of the test points listed in table IV. The pressure shall be decreased to bring the pointer approximately to the desired reading, then held constant while two readings are taken, the first before the altimeter is tapped and the second after tapping. The difference between any two such readings shall be recorded as friction and shall not exceed the tolerances listed in table IV. This test may be combined with the tests specified in 4.6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

TABLE IV. Friction.

Altitude Feet	Tolerance ±Feet
1,000	70
2,000	70
3,000	70
4,000	80
10,000	90
16,000	100
20,000	115
25,000	130
30,000	150
35,000	175
40,000	200
50,000	250
60,000	450

4.6.8 Case leakage. A vacuum shall be slowly applied to the static pressure port to produce an indication of approximately 40,000 feet. The vacuum line shall then be valved or pinched off. After 1 minute, the indication shall not have decreased by more than 100 feet.

4.6.9 Magnetic effect. The altimeter, nonoperating, shall be rotated in a vertical plane about a short-bar-magnet compass with the nearest part of the indicator 5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 1°. The test shall be repeated with the altimeter at rated voltage.

4.6.10 High temperature. The altimeter shall be subjected to a temperature of +65° ±2°C for a period of 3 hours. It shall be allowed to cool to normal temperature, after which it shall be subjected to the tests specified in 4.6.3 and shall meet the requirements specified. During and after the test, the setting system shall operate properly.

4.6.11 Low temperature exposure. This test shall be conducted in accordance with procedure II of MIL-E-5272. After

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

completion of the test, the altimeter shall be subjected to and shall meet the requirements specified in 4.6.3.

4.6.12 High temperature exposure. This test shall be conducted in accordance with procedure II, high temperature test, of MIL-E-5272 except that the temperature shall be +65° ±2°C. After the indicator has returned to room temperature, it shall meet the requirements for room temperature scale error.

4.6.13 Humidity. The indicator shall be subjected to a humidity test in accordance with procedure I of MIL-E-5272. Within 1 hour after the 360-hour period, the indicator shall meet the requirements of the Scale error at room temperature inspection.

4.6.14 Salt spray. The indicator shall be subjected to requirements of procedure I of MIL-E-5272 for a period of 50 hours.

4.6.15 Vibration error. The altimeter shall be mounted on a vibration stand with a line joining the centers of the two upper mounting holes in a horizontal plane and with the plane of the mounting lugs of the indicator vertical. The altimeter shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3000 cpm to determine that the natural frequencies of the indicator do not occur in this frequency range. While the indicator is vibrated, the maximum amplitudes of pointer oscillation shall not exceed 20 feet and the pointer variation from its original position shall not exceed 20 feet.

4.6.16 Vibration failure. The altimeter shall be subjected to procedure V of MIL-E-5272. The indicator shall be mounted on the vibration stand with a line joining the centers of the two upper mounting holes horizontal and the dial in a vertical plane. The static pressure port shall be connected to a vacuum source and to a reference mercury barometer. The vacuum shall be adjusted to produce any desired altitude indication. The indicator shall be vibrated with a circular motion, 0.018 to 0.020 inch in diameter, in a plane inclined 45° to the horizontal, and the frequency shall be varied uniformly from 5 to 50 cps and return, once each hour for a 3-hour period. Following the vibration period, the altimeter shall be subjected to and shall meet the requirements specified in 4.6.3.

4.6.17 Fungus. The indicator shall be subjected to fungus resistance test in accordance with procedure I of MIL-E-5272.

4.6.18 Rain. The altimeter shall be subjected to a rain test in accordance with procedure II of MIL-E-5272.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

4.6.19 Sand and dust. The altimeter shall be subjected to a sand and dust test in accordance with procedure I of MIL-E-5272.

4.6.20 Acceleration. The altimeter shall be subjected to procedure I of MIL-E-5272 except that the altimeter shall be mounted in its normal operation with the rear of the case toward the center of rotation and shall be subjected to an acceleration of 20g for a period of 30 seconds. This test shall be repeated with the altimeter mounted successively in the following positions:

a. Rotated 90° in azimuth with respect to the original position.

b. Dial horizontal, with the pointer toward the center of rotation.

At the end of this acceleration, the altimeter shall meet the individual tests except for scale error at -35°C.

4.6.21 Endurance, stability and shelf life. The altimeter shall be subjected to and shall meet the requirements of the Individual inspections. The Scale error at room temperature readings will be referred to later in this test at "the original scale error readings." The instrument shall be subjected to a pressure cycling process consisting of 2000 applications of an absolute pressure variation equivalent to approximately 50,000 feet pressure altitude. After 500, 1000, and 1500 cycles, the altimeter shall be subjected to 2 cycles of temperature-shock, procedure I of MIL-E-5272, except that the high temperature shall be +71°C in lieu of +85°C. Each pressure cycle shall consist of decreasing the pressure at a rate of 6000 ±2000 fpm until 50,000 ±5000 feet is reached. The pressure shall be maintained at this pressure altitude for no less than 2 minutes. The pressure shall then be increased at a rate of 6000 ±2000 fpm until ambient pressure altitude is reached. After a minimum of 2 minutes at ambient pressure, the next cycle shall be commenced. This shall be continued until 2000 cycles have been completed; after which, the altimeter shall rest at ambient pressure for approximately 48 hours. The contractor may remove the altimeter at any desired time during the cycling process to determine its performance. Following completion of all cycling and the 48-hour rest period, the altimeter may be reset by means of the zero-setting adjustment mechanism. The mount of reset shall be limited to 100 feet. The instruments shall then be subjected to and shall meet the requirements of the Individual inspections. In addition, the Scale errors at room temperature shall not deviate from the original scale error readings by more than the values specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

TABLE V. Scale error variation.

Standard Altitude Feet	Scale Error Variation Tolerances Feet
0	30
500	30
1,000	30
1,500	35
2,000	35
2,500	35
3,000	35
4,000	40
5,000	45
6,000	45
8,000	50
10,000	55
12,000	60
14,000	65
15,000	70
16,000	70
18,000	75
20,000	80
22,000	85
25,000	90
30,000	105
35,000	120
40,000	130
45,000	140
50,000	155
55,000	180
60,000	405
70,000	655
80,000	900

Following completion of the above tests, the altimeter shall be placed in storage for no less than 30 days. During the 30-day period, the instrument shall remain as dormant as practicable. The mechanism shall not be exercised other than that caused by normal ambient barometric pressure variations. The temperature shall remain essentially constant; however, normal room temperature variations are permissible. After the 30-day period, the altimeter shall again be subjected to and shall meet the requirements of the Individual inspections. In addition to meeting the tolerances specified in table II for scale error, the readings shall not vary from the original scale error readings by more than the values specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

4.6.22 Integral lighting. The altimeter shall be subjected to all the tests under test methods of MIL-L-25467.

4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in section 4 of MIL-P-115, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with method 1A8 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.7 of this specification.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box shall be used for the shipping container.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping container shall conform to specification PPP-B-636, class weather-resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The altimeter covered by this specification is intended for use as a standby indicator in high performance aircraft to give accurate indications of altitude.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing desired, and reinspection date markings.

c. Whether sampling plan B tests are to be conducted.

6.3 Definitions.

6.3.1 Tapping. The phrase "tapped lightly" is defined as meaning the application of a light vibration sufficient to over-

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

come that amount of residual friction normally associated with high quality aircraft instruments.

6.4 Crash provisions. The requirements specified in 3.5 are intended to facilitate aircraft crash investigations by minor additions to standard indicator and transmitter techniques. Complex mechanisms to meet the requirements are not desired. Techniques such as fluorescent-luminescent paint on back side of pointer, index marks on internal moving parts, impalement spikes, or like accessories, are preferred.

6.5 First article. A first article is required for inspection and approval (see 3.1 and 4.4).

6.6 International standardization agreement. Certain provisions (see paragraph 3.9) of this specification are the subject of international standardization agreement STANAG 3433. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Revision changes. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

Custodian:
Air Force - 99

Reviewer:
Air Force - 99

Preparing Activity:
Air Force - 99

Project Number:
6610-F299

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-A-27391B

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:39Z
Check the source to verify that this is the current version before use.